ONE-PAGE BRIEF

Post-quantum cryptography review: canton @ 4eab7a047040

Client: Digital Asset (Canton) — public repository (passive review) · Source: github.com/digital-asset/canton, commit `4eab7a047040` · Reviewed: 2026-11-06 15:24 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 2 of 4

post-quantum in use outside the critical paths

FINDINGS

0 Critical **4** High **9** Medium **4** Low **0** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Extend post-quantum protection to the critical roles that stay classical, starting with what can be recorded today (wrapped keys, channel key exchange).

WHAT MATTERS MOST

High QSI-2026-19406	**Private key material present in performance/src/main/console/tls/participant.pem** What to do: Treat the key as compromised: rotate it, revoke associated certificates, remove it from version history and store keys in a secrets...
High QSI-2026-19410	**Private key material present in performance/src/main/console/tls/synchronizer.pem** What to do: Treat the key as compromised: rotate it, revoke associated certificates, remove it from version history and store keys in a secrets...
High QSI-2026-19403	**Certificate has expired (localhost)** What to do: Renew the certificate and automate renewal (e.g. ACME) with expiry monitoring.
High QSI-2026-19407	**Certificate has expired (localhost)** What to do: Renew the certificate and automate renewal (e.g. ACME) with expiry monitoring.

13 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium key wrapping and data encryption at rest	ECDH	Wrap keys with ML-KEM or a hybrid KEM (X25519 + ML-KEM-768) and re-wrap the stored keys; data that may...
Medium transaction and message signing	ECDSA, Ed25519, RSA	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium authentication and identity (certificates, tokens, device attestation)	RSA, ECDSA, Ed25519	Keep credentials short-lived and plan ML-DSA certificates, tokens and keys as the ecosystem supports them
Medium role not determined	ECDSA, RSA	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

Static review of source code and manifests; files analysed: 3314. No requests were sent to the owner's systems.

Full report, content SHA-256: `efbd8a018afeed64113c4427d45e8d0033b998dc55ff0d735a440be80e9c9cb1`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.